

# Circular Economy Ontology Network (CEON) - Product Module

## Metadata

### IRI

<http://w3id.org/CEON/ontology/product/>

### Title

Circular Economy Ontology Network (CEON) - Product Module

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### License

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### Version Iri

<http://w3id.org/CEON/ontology/product/0.4/>

### Version Info

0.4

### Preferred Namespace Prefix

product

### Preferred Namespace Uri

<http://w3id.org/CEON/ontology/product/>

### Description

The Product module of CEON (Circular Economy Ontology Network).

### Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-2, CVN-ResourceType-4, C11-2, C12-1, C13-3, E2-1, T8-3.

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.2: CE5-2, C11-2, C12-1, C13-3, E2-1, T8-3.

## Classes

### Actor<sup>C</sup>

#### IRI

<http://w3id.org/CEON/ontology/actorODP/Actor>

### Resource Relation<sup>C</sup>

#### IRI

<http://w3id.org/CEON/ontology/actorODP/ResourceRelation>

#### Super Class Of

[resourceODP:Composition<sup>C</sup>](#)

## Process<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/processODP/Process>

## Compliance<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/Compliance>

**In Domain Of** [complianceWith](#)<sup>op</sup>  
[hasCertificate](#)<sup>dp</sup>

**In Range Of** [hasCompliance](#)<sup>op</sup>

## Item<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/Item>

**Description** An item is a physical object put into a market for sale, i.e. corresponding to an individual of a product. The item can be a thing that grows naturally or produced through some chemical or manufacturing processes.

**Sub Class Of** [resourceODP:PhysicalObject](#)<sup>C</sup>

**In Domain Of** [hasProductObjectComponent](#)<sup>op</sup>  
[modelled by](#)<sup>op</sup>

**In Range Of** [associatedWithProductObject](#)<sup>op</sup>  
[hasProductObjectComponent](#)<sup>op</sup>

**Restriction** [modelled by](#)<sup>op</sup> *exactly* 1 [Item](#)<sup>C</sup>

## Matter composition<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/MatterComposition>

**Description** A product composition is the composed information of a product in terms of a specific component, in which such a component is also a matter.

**Sub Class Of** [resourceODP:Composition](#)<sup>C</sup>

**In Domain Of** [associated with matter](#)<sup>op</sup>

**Restriction** [associated with matter](#)<sup>op</sup> *exactly* 1 [Matter composition](#)<sup>C</sup>

## Product<sup>C</sup>

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                  | <a href="http://w3id.org/CE0N/ontology/product/Product">http://w3id.org/CE0N/ontology/product/Product</a>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b><u>Is Defined By</u></b> | ISO 59004:2024 -3.2.2 product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b><u>Description</u></b>   | <p>Physical-based object designed for or utilized with a purpose. A product can be, for example: goods of any type; hardware (e.g. engine mechanical part, spare parts, consumables); electrical or electronic hardware devices or components (e.g. computers, communication equipment and sensors); processed materials (e.g. lubricant, cement).</p> <p>Represents the common sense notion of a product, i.e. the abstract notion of a product type, which is a model that is used for manufacturing items of that product.</p> |
| <b><u>Sub Class Of</u></b>  | <a href="#">Solution<sup>C</sup></a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b><u>In Domain Of</u></b>  | <a href="#">has product component</a> <sup>OP</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b><u>In Range Of</u></b>   | <a href="#">associated with product model</a> <sup>OP</sup><br><a href="#">batch of products</a> <sup>OP</sup><br><a href="#">has product component</a> <sup>OP</sup><br><a href="#">modelled by</a> <sup>OP</sup>                                                                                                                                                                                                                                                                                                                |
| <b><u>Restriction</u></b>   | <a href="#">hasComposition</a> <sup>OP</sup> some <a href="#">Product<sup>C</sup></a><br><a href="#">hasComposition</a> <sup>OP</sup> some <a href="#">Product<sup>C</sup></a>                                                                                                                                                                                                                                                                                                                                                    |

## Product composition<sup>C</sup>

|                            |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/product/ProductComposition">http://w3id.org/CE0N/ontology/product/ProductComposition</a>               |
| <b><u>Description</u></b>  | A product composition is the composed information of a product in terms of a specific component, in which such a component is also a product. |
| <b><u>Sub Class Of</u></b> | <a href="#">resourceODP:Composition<sup>C</sup></a>                                                                                           |
| <b><u>In Domain Of</u></b> | <a href="#">associated with product model</a> <sup>OP</sup>                                                                                   |
| <b><u>Restriction</u></b>  | <a href="#">associated with product model</a> <sup>OP</sup> <i>exactly</i> 1 <a href="#">Product composition<sup>C</sup></a>                  |

## Product Object Composition<sup>C</sup>

|                            |                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/product/ProductObjectComposition">http://w3id.org/CE0N/ontology/product/ProductObjectComposition</a> |
| <b><u>Sub Class Of</u></b> | <a href="#">resourceODP:Composition<sup>C</sup></a>                                                                                         |
| <b><u>In Domain Of</u></b> | <a href="#">associatedWithProductObject</a> <sup>OP</sup>                                                                                   |
| <b><u>Restriction</u></b>  | <a href="#">associatedWithProductObject</a> <sup>OP</sup> <i>exactly</i> 1 <a href="#">ProductObjectComposition<sup>C</sup></a>             |

## Solution<sup>C</sup>

|                              |                                                                                                             |
|------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/product/Solution">http://w3id.org/CE0N/ontology/product/Solution</a> |
| <b><u>Is Defined By</u></b>  | ISO 59004:2024 - 3.2.1 solution                                                                             |
| <b><u>Description</u></b>    | Product or service, or a combination thereof, that fulfils a need of an interested party.                   |
| <b><u>Super Class Of</u></b> | <a href="#">Product<sup>C</sup></a>                                                                         |

## Batch Of Objects<sup>C</sup>

|                            |                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                 | <a href="http://w3id.org/CE0N/ontology/resourceODP/BatchOfObjects">http://w3id.org/CE0N/ontology/resourceODP/BatchOfObjects</a> |
| <b><u>In Domain Of</u></b> | <a href="#">batch of products<sup>op</sup></a>                                                                                  |
| <b><u>Restriction</u></b>  | <a href="#">batch of products<sup>op</sup></a> <i>exactly</i> 1 <a href="#">resourceODP:BatchOfObjects<sup>C</sup></a>          |

## Composition<sup>C</sup>

|                              |                                                                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/resourceODP/Composition">http://w3id.org/CE0N/ontology/resourceODP/Composition</a>                                 |
| <b><u>Sub Class Of</u></b>   | <a href="#">actorODP:ResourceRelation<sup>C</sup></a>                                                                                                     |
| <b><u>In Domain Of</u></b>   | <a href="#">compositionOf<sup>op</sup></a><br><a href="#">Composition quantity<sup>dp</sup></a>                                                           |
| <b><u>In Range Of</u></b>    | <a href="#">hasComposition<sup>op</sup></a>                                                                                                               |
| <b><u>Super Class Of</u></b> | <a href="#">Matter composition<sup>C</sup></a><br><a href="#">Product composition<sup>C</sup></a><br><a href="#">ProductObjectComposition<sup>C</sup></a> |

## Matter<sup>C</sup>

|                           |                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                | <a href="http://w3id.org/CE0N/ontology/resourceODP/Matter">http://w3id.org/CE0N/ontology/resourceODP/Matter</a> |
| <b><u>In Range Of</u></b> | <a href="#">associated with matter<sup>op</sup></a>                                                             |

## Physical Object<sup>C</sup>

|                              |                                                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>                   | <a href="http://w3id.org/CE0N/ontology/resourceODP/PhysicalObject">http://w3id.org/CE0N/ontology/resourceODP/PhysicalObject</a> |
| <b><u>Super Class Of</u></b> | <a href="#">Item<sup>C</sup></a>                                                                                                |

## Object Properties

### associated with matter<sup>op</sup>

|                    |                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/product/associatedWithMatter">http://w3id.org/CE0N/ontology/product/associatedWithMatter</a> |
| <b>Description</b> | associatedWithMatter intends to represent the matter to which a matter composition information regards to.                          |
| <b>Domain</b>      | <a href="#">Matter composition</a> <sup>c</sup>                                                                                     |
| <b>Range</b>       | <a href="#">resourceODP:Matter</a> <sup>c</sup>                                                                                     |

### associated with product model<sup>op</sup>

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/product/associatedWithProductModel">http://w3id.org/CE0N/ontology/product/associatedWithProductModel</a> |
| <b>Description</b> | associatedWithProductModel intends to represent the product model to which a product composition information regards to.                        |
| <b>Domain</b>      | <a href="#">Product composition</a> <sup>c</sup>                                                                                                |
| <b>Range</b>       | <a href="#">Product</a> <sup>c</sup>                                                                                                            |

### associated with product object<sup>op</sup>

|                        |                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/product/associatedWithProductObject">http://w3id.org/CE0N/ontology/product/associatedWithProductObject</a> |
| <b>Sub Property Of</b> | <a href="#">topObjectProperty</a> <sup>op</sup>                                                                                                   |
| <b>Domain</b>          | <a href="#">ProductObjectComposition</a> <sup>c</sup>                                                                                             |
| <b>Range</b>           | <a href="#">Item</a> <sup>c</sup>                                                                                                                 |

### batch of products<sup>op</sup>

|                    |                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/product/batchOfProduct">http://w3id.org/CE0N/ontology/product/batchOfProduct</a> |
| <b>Description</b> | batchOfProduct intends to represent what is the model product of a batch of objects.                                    |
| <b>Domain</b>      | <a href="#">resourceODP:BatchOfObjects</a> <sup>c</sup>                                                                 |
| <b>Range</b>       | <a href="#">Product</a> <sup>c</sup>                                                                                    |

### compliance with<sup>op</sup>

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/product/complianceWith">http://w3id.org/CE0N/ontology/product/complianceWith</a> |
| <b>Domain</b> | <a href="#">Compliance</a> <sup>c</sup>                                                                                 |

## composition of<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/compositionOf>

**Domain** [resourceODP:Composition](#)<sup>C</sup>

**Range** [resourceODP:Matter](#)<sup>C</sup> or [Product](#)<sup>C</sup> or [Item](#)<sup>C</sup>

## has compliance<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/hasCompliance>

**Domain** [Product](#)<sup>C</sup> or <http://w3id.org/CE0N/ontology/processODP/Process><sup>C</sup> or [Item](#)<sup>C</sup> or [actorODP:Actor](#)<sup>C</sup>

**Range** [Compliance](#)<sup>C</sup>

## has composition<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/hasComposition>

**Description** hasComposition intends to represent that a product model or physical object can have composition information.

**Domain** [Product](#)<sup>C</sup> or [resourceODP:PhysicalObject](#)<sup>C</sup>

**Range** [resourceODP:Composition](#)<sup>C</sup>

## has product component<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/hasProductComponent>

**Description** hasProductComponent intends to represent that a product can have other product components.

**Domain** [Product](#)<sup>C</sup>

**Range** [Product](#)<sup>C</sup>

## has product object component<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/hasProductObjectComponent>

**Domain** [Item](#)<sup>C</sup>

**Range** [Item](#)<sup>C</sup>

modelled by<sup>dp</sup>

|                    |                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/product/modelledBy">http://w3id.org/CE0N/ontology/product/modelledBy</a> |
| <b>Description</b> | modelledBy intends to represent that product objects follow particular product models.                          |
| <b>Domain</b>      | <a href="#">Item</a> <sup>c</sup>                                                                               |
| <b>Range</b>       | <a href="#">Product</a> <sup>c</sup>                                                                            |

## Datatype Properties

Composition quantity<sup>dp</sup>

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/product/compositionQuantity">http://w3id.org/CE0N/ontology/product/compositionQuantity</a> |
| <b>Domain</b> | <a href="#">resourceODP:Composition</a> <sup>c</sup>                                                                              |
| <b>Range</b>  | <a href="#">xsd:double</a>                                                                                                        |

country of assembly<sup>dp</sup>

|                        |                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/product/countryOfAssembly">http://w3id.org/CE0N/ontology/product/countryOfAssembly</a> |
| <b>Sub Property Of</b> | <a href="#">locationOfAssembly</a> <sup>dp</sup>                                                                              |

country of manufacture<sup>dp</sup>

|                        |                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/product/countryOfManufacture">http://w3id.org/CE0N/ontology/product/countryOfManufacture</a> |
| <b>Sub Property Of</b> | <a href="#">locationOfManufacture</a> <sup>dp</sup>                                                                                 |

country of origin<sup>dp</sup>

|                        |                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>             | <a href="http://w3id.org/CE0N/ontology/product/countryOfOrigin">http://w3id.org/CE0N/ontology/product/countryOfOrigin</a> |
| <b>Sub Property Of</b> | <a href="#">locationOfOrigin</a> <sup>dp</sup>                                                                            |

has certificate<sup>dp</sup>

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>    | <a href="http://w3id.org/CE0N/ontology/product/hasCertificate">http://w3id.org/CE0N/ontology/product/hasCertificate</a> |
| <b>Domain</b> | <a href="#">Compliance</a> <sup>c</sup>                                                                                 |

location of assembly<sup>dp</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/locationOfAssembly>

**Super Property Of** [countryOfAssembly](#)<sup>dp</sup>

**Domain** [Item](#)<sup>c</sup> or [Product](#)<sup>c</sup>

**Range** [xsd:string](#)

location of manufacture<sup>dp</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/locationOfManufacture>

**Super Property Of** [countryOfManufacture](#)<sup>dp</sup>

**Domain** [Item](#)<sup>c</sup> or [Product](#)<sup>c</sup>

**Range** [xsd:string](#)

location of origin<sup>dp</sup>

**IRI** <http://w3id.org/CE0N/ontology/product/locationOfOrigin>

**Super Property Of** [countryOfOrigin](#)<sup>dp</sup>

**Domain** [Product](#)<sup>c</sup> or [Item](#)<sup>c</sup>

**Range** [xsd:string](#)

## Annotation Properties

contributor<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/contributor>

created<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/created>

creator<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/creator>

description<sup>ap</sup>

**IRI** <http://purl.org/dc/terms/description>

|                                          |                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| license <sup>ap</sup>                    |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/dc/terms/license">http://purl.org/dc/terms/license</a>                                                                                                               |
| title <sup>ap</sup>                      |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/dc/terms/title">http://purl.org/dc/terms/title</a>                                                                                                                   |
| preferred namespace prefix <sup>ap</sup> |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespacePrefix">http://purl.org/vocab/vann/preferredNamespacePrefix</a>                                                                         |
| preferred namespace uri <sup>ap</sup>    |                                                                                                                                                                                               |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespaceUri">http://purl.org/vocab/vann/preferredNamespaceUri</a>                                                                               |
| covers requirements <sup>ap</sup>        |                                                                                                                                                                                               |
| IRI                                      | <a href="http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements">http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements</a> |

## Namespaces

:  
<http://w3id.org/CE0N/ontology/product/>  
 actorODP  
<http://w3id.org/CE0N/ontology/actorODP/>  
 dcterms  
<http://purl.org/dc/terms/>  
 odp  
<http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#>  
 owl  
<http://www.w3.org/2002/07/owl#>  
 prov  
<http://www.w3.org/ns/prov#>  
 rdf  
<http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
 rdfs  
<http://www.w3.org/2000/01/rdf-schema#>  
 resourceODP  
<http://w3id.org/CE0N/ontology/resourceODP/>  
 vann  
<http://purl.org/vocab/vann/>  
 xsd  
<http://www.w3.org/2001/XMLSchema#>

## Legend

|    |                          |
|----|--------------------------|
| c  | Classes                  |
| op | Object<br>Properties     |
| dp | Datatype<br>Properties   |
| ap | Annotation<br>Properties |